Investigation into the usability of Games Console dashboards/in game menu screens

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Abstract:

With the current generation of games consoles, there appears to be a lack of development into the usability of these console's dashboards. Trends, such as features being hidden so that new users cannot access key areas, are prevalent across these dashboards. This research project investigated the current generation of games console dashboards and gauged how users feel about these dashboards using questionnaires and usability studies. The research showed many issues facing current consoles, and these issues were used to test newly designed dashboards against participants. A questionnaire was first used to gather general opinions from participants which helped to decide on what areas needed testing for a usability study involving two of the current generation consoles. The results of these studies helped to design three prototype dashboards that were made and tested against participants to test differing ideas towards a dashboard's design and to see which areas should be included in a final dashboard. With these tests, a final dashboard was created to tackle the main issues that face current dashboards and how UI designers could tailor dashboards in the future for both new and existing users.

Chapter 1 - Introduction:

1.1 - Project Background

In the past few decades, modern video games consoles have seen a surge in popularity with the current generation selling tens of millions of consoles every year and that number rising every year (Sherif, 2024). Despite the increasing popularity of games consoles and the importance of the dashboard in the user experience, it appears that there is a lack of research into the usability of games console dashboards (see Chapter 2, Literature Review). It is likely that the games console dashboards are designed without reference to usability design guidelines and research that leads to a lack of usability design or inadequate visual design. This project focuses on research into usability regarding graphical user interfaces and how to apply that knowledge onto a games console's dashboard. This will be partly conducted through usability tests with participants as a way of direct feedback between a user and the dashboards. One of the main issues that has arisen with current generation consoles is that there is an obvious lack of visual and usability design put into the dashboards of the consoles. Current consoles have to display many icons and that information needing to be displayed on a dashboard at one time can intrude on its usability aspects, which can confuse users or turn them away from continuing using the consoles. A lot of research into usability was done by Jakob Nielsen, who invented the concept of heuristic evaluation and developed a set of ten heuristics that are widely used in development of user interfaces (Nielsen, 1994). With Nielsen having a large influence on usability frameworks and the principles that go with them, a lot of the study will be based around his research and the set of heuristics that were published as they are still the most used framework for the development of a user interface.

1.2 - Project Objectives

The main objective of the project will be to determine what are considered to be the positive and negative aspects of a games console dashboard regarding usability and visual design elements. This information will be determined through the use of participant studies and the findings will then be used to create a games console dashboard that is designed with usability and visual design in mind for users.

- Develop and send out a questionnaire to gather opinions about the latest console dashboards regarding usability and visual design.
- Conduct a usability study using participants to collect data about their ability to perform tasks on current generation games console dashboards.
- Develop a set of prototype dashboards from the data gathered from the questionnaire and usability study that will be used to gather comments about the dashboard's usability and design elements.
- With the feedback from the prototype dashboards, develop a final dashboard that incorporates features from all the information gathered.

Chapter 2 - Literature Review:

Since 1972 with the Magnavox Odyssey, the first home video games console (Burke, 2023), video games have been able to be directly accessed from a user's home. Since then, video game consoles have sold hundreds of millions across the years.

Sherif (2024) shows a graph containing the statistics for the unit sales amount of current generation consoles from 2017 to 2023. The statistics show that the PS5 and Nintendo Switch were the highest selling consoles from 2021 to 2023, with the PS5 selling 21.8 million units in 2023 and the Switch selling 16.4 million units in 2023. As such, these are consoles that users are the most likely to have experience with and can be used as control dashboards for when testing and making games console dashboards.

Nielsen (1994) introduced ten general principles that can be used for interaction design. These "heuristics" are made to be broad rules to follow and are not guidelines but can help lead designers into the right direction (See Figure 1).

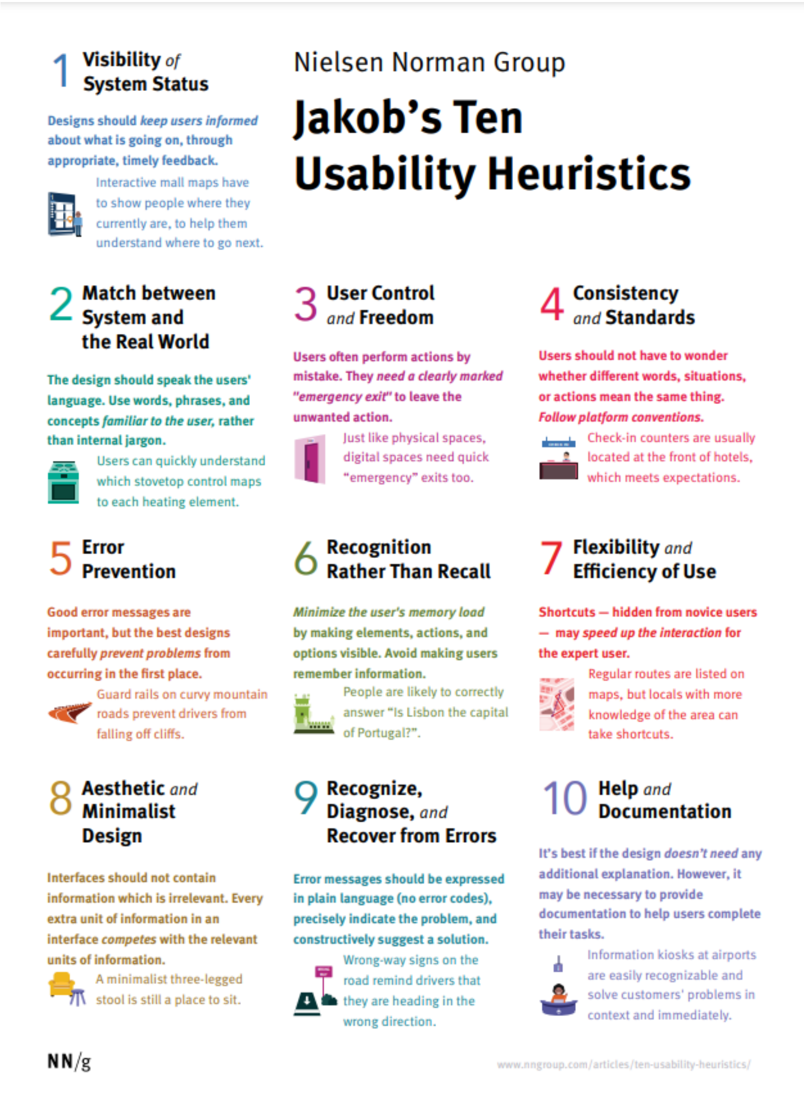


Figure 1: a poster showcasing all ten of Nielsen's usability heuristics (Nielsen, 1994)

An example of one of the ten heuristics is "Visibility of System Status". This principle advises designers that the interface should always keep users informed about what is going on. This can be done with appropriate feedback within a suitable amount of time. The constant feedback towards users, with it appearing immediately whenever possible, helps them to understand what their actions have done and how they can process what their next steps should be. For a games console dashboard, this would be making sure that the user constantly knows what icons they have selected or having a label that shows what area of the dashboard a user is in. The PlayStation 5 (PS5) dashboard follows this principle as the area of the dashboard the user is currently in is highlighted in bold white (see Figure 2).

Figure 2: Showcase of a label describing the area the user is in on the PS5 dashboard

The whole set of heuristics created by Nielsen is important as it provides clear areas to focus on while testing dashboards for their usability. Those areas can then be tested specifically during participant studies to see if the dashboards are following the principles.

Glinert (2009) wrote about “4 dimensions of usability” that can create a good user interface. These being learnability, simplicity, efficiency and aesthetic. Learnability follows the same concept as Nielsen’s heuristic Recognition rather than Recall where a user should be able to access any area of the dashboard purely through what is shown to them, over what they remember being taught. The element of learnability expands on this by having trial and error be a main factor with feedback as a main contributor to learning the dashboard. A user might make mistakes that they can learn from and, if they do make mistakes, they should be able to easily undo their actions. This requires straightforward ways to return from the menu or area the user is in on a dashboard, with either a back button shown in the area or having a prompt of what button to press on a controller. Efficiency will concern the issue of how quickly a task can be completed. For a dashboard, having lots of icons can clutter the dashboard and make it so that a user will have scroll through multiple icons to get to the area they want to enter. An efficient dashboard will allow its users to quickly accomplish any task they want to complete. For a console like the PS5, this issue was addressed by having two areas for the games and the media apps so that a user does not have to scroll through multiple games icons to access a media application. However, for a console like the Xbox One, every icon is in one singular area, as shown in Figure 3.

This can be confusing for a new user to access and navigate due to the amount of information on a singular page.

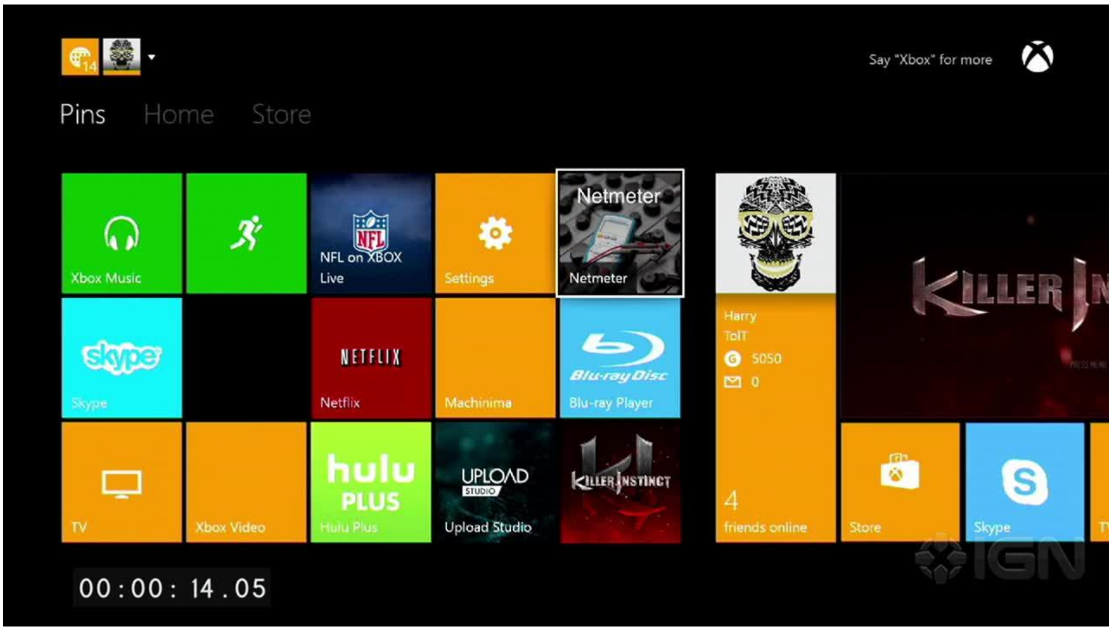
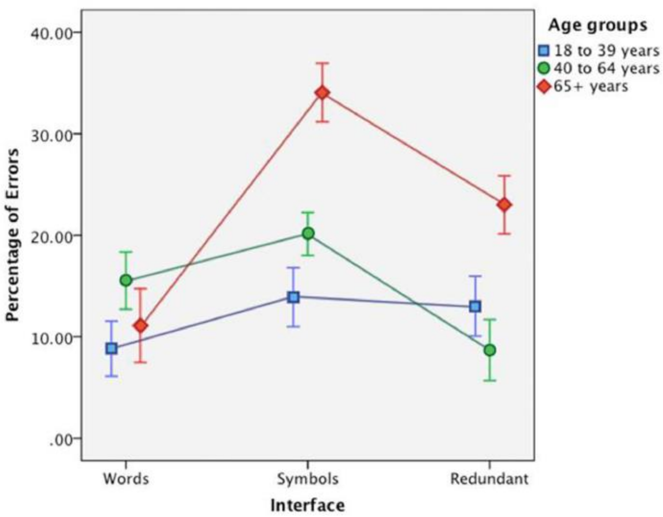


Figure 3: Showcase of the Xbox One dashboard

Simplicity is the concept of making sure that all the controls for a user are as simple as possible for them to understand. For a games console dashboard, this will involve making sure that every area on the dashboard is easy to understand and that a user can understand every element shown to them on the dashboard when handed a controller. This can contrast with efficiency because making a dashboard easy to understand can cause the dashboard to be slower to navigate due to needing to keep every aspect simple over making them efficient. Aesthetic is the largest aspect when considering the dimensions for a games console's dashboard. The dashboard must be visually distinct and pleasing for a user, so that it is enjoyable to use and to ensure that the dashboard can follow the other three dimensions. While these dimensions are good guidelines for designing dashboards, it is not easy to follow all 4 of them and improve all of them collectively. Improving the simplicity of a dashboard may involve removing some elements that improve the efficiency of the dashboard. In the case of a games console dashboard, it is important to focus mainly on the aesthetics and simplicity of the dashboard over the efficiency and learnability, however making sure every dimension is included should be completed when possible.

Reddy et Al. (2019) performed an experiment involving groups of participants about different types of user interfaces to see which were best for differing age groups. The participants were split into 18 to 39, 40 to 64 and 65+ age groups, with tasks being performed on separate user interfaces. The user interfaces were words only, icons only and a redundant group that combined the two to see which groups reacted best with each specific dashboard. The results of the experiments showed that while participants aged 65 and older were more likely make mistakes or need help with their interfaces, the interface they were best with included only words. The figures below show the full results regarding errors and show that both participant groups from 18 to 64 performed the best when given an interface with both text and icons.



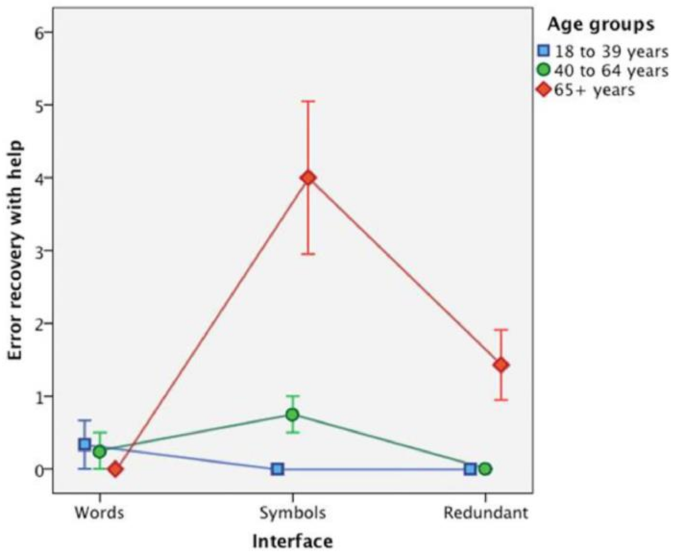


Figure 4: Two graphs that show the results regarding errors for all participant groups in the experiment performed by Reddy et al. (2019).

The overall results can be used for designing the dashboards as it shows that the best type of user interface for most ages will include both icons and text. However, for designing the dashboards to accommodate for older adults, the dashboard can include more text-based areas on specific pages. This will mean that older adults trying to use the dashboards are not turned away from using the dashboards and are properly designed for.

Chapter 3 - Methodology:

3.1 - The Waterfall Model

The primary software development methodology chosen for the project is the waterfall model (Hoory, Bottorff and Watts, 2024). The model was chosen due to its sequential approach whereby each task only started once the previous task had been fully completed as it is most appropriate for this project. For example, the research must be completed before the design of the initial dashboards, and the studies with dashboards must be completed before designing the final prototype dashboard.

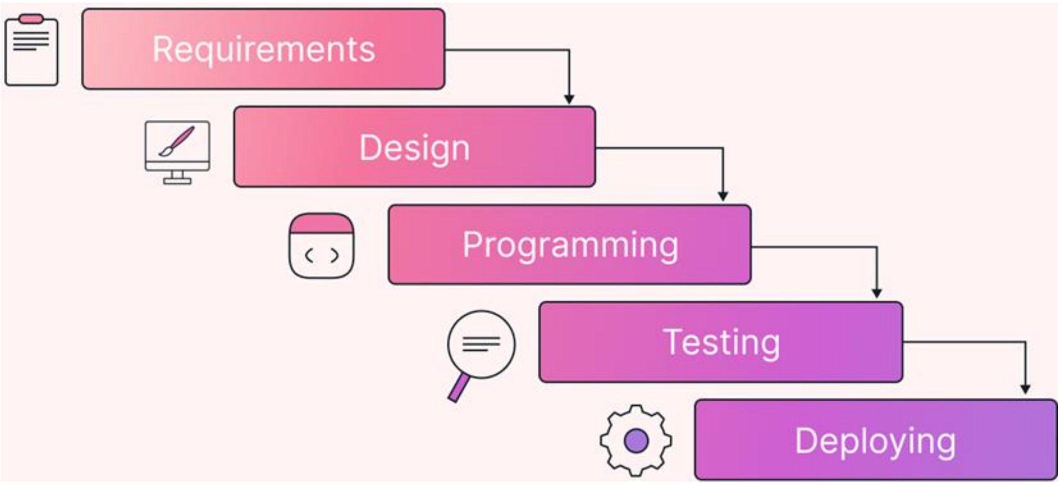


Figure 5: A diagram showing the stages that will be used from the waterwall model.

Another advantage of the waterfall model is that because every step must be fully completed before proceeding to the next step, it means that all areas that need to be covered during the stage can be finished, and all the results can be used in the next stage. This is more relevant than the agile methodology (Gurnov, 2025) for a project such as this as it means there is no need to go back and forth between the participant studies while creating the prototype dashboards because there will be no more feedback given. The fixed step count for the methodology also means that the goals to reach the end product are hard to deviate from, which allows the step currently being completed to have full attention. The attention towards each step will allow for less detail to be lost when compared to other methodologies, where focus could be lost on what is needed for the step. However, an issue with the waterfall model is that not being able to return to an earlier step means that changes can be difficult to make. This could be an issue for this project if, for example, it was discovered that more information was required only after the questionnaires had been completed. Strictly following the waterfall model would mean that it would not be possible to return to the questionnaire stage to gather more information once the usability studies started to be performed. Another problem with the waterfall model is that deciding what steps are necessary can be an issue when designating what the project will need. As the steps to complete need to be made before the project is started, this means that the necessary feedback from users to make a complete final dashboard must be assumed. This can lead to issues during the project as it means that crucial steps can be missed out and cannot be returned to, so they can be fixed. Despite these issues, the waterfall methodology is the best for the project, as each step needs the full feedback from users to continue and to know what to test for the upcoming step.

3.2 - Qualitative Research

The primary research type chose for the participant studies was qualitative research (Bhandari, 2025). With this research type, participants can give direct feedback about their thoughts regarding specific dashboards. This gives a larger range of answers, compared to the quantitative approach (Sreekumar, 2023) which would focus on getting participants to rate areas of the dashboard on a scale if used. Qualitative research means that participants can directly talk about their experiences or about the decisions they have made, which can help with future design choices. This was the main research type that the participant studies were based around, as allowing the participants to verbally explain their actions provides insight into what the main audience will think about when they are in control of the dashboards. The amount of data given during the studies will be categorized and put into groups using thematic analysis (Rosala, 2022). Thematic analysis was chosen to help split the qualitative data into separate categories to help, as to analyse them better. Thematic analysis will help with finding specific areas that are issues as the grouping of data into categories can show what the most problematic areas are, and they can be addressed accordingly. An issue with deciding to do thematic analysis is that with a large pool of data, it can be very time consuming to read through it all and then categorise it. However, categorising the data will not be hard as setting broad categories for the analysis means that, in combination with the shorter form answers, finding the specific comments required was not too difficult.

3.3 - Questionnaire

A questionnaire was chosen as the first study done during the project, due to it being a low-cost method that can result in large quantities of data produced (Ranganathan and Caduff, 2023). The main strength of the questionnaire is the large area that data can be gathered from. Google forms was the main software for the questionnaire due to it being an online service, further increasing the range of participants the questionnaire reached. The questionnaire was designed so that all the questions were open ended, which encourages participants to give a complete answer with justifications. These questions led to more depth in participants answers, as the problems with each dashboard were highlighted and the reasons for them being problems were also written. However, due to using open ended questions for the questionnaire, participants are less inclined to speak about further issues outside of the strict range of questions. However, such information could then be gained during the following usability studies using techniques such as Think Aloud (Nielsen, 2012) that encourage participants to expand upon their answers in more depth. An issue that can also arise for the questionnaire is a misunderstanding of the questions given. For example, a question about the visual design of a console could be interpreted as talking about the layout of the dashboard in terms of the icon's placement and size. However, another participant could interpret that question as asking about the colours and background of the dashboard. This can lead to issues where the question's original intent is not answered properly and therefore, cannot be used for design purposes as there is no usable information given. This can be avoided by making the questions clear about what they are asking.

3.4 - Usability Testing

For the participant studies, usability testing (Moran, 2019) will be used to assess the Nintendo Switch and PS5 dashboards and to see how they perform with participants. Usability testing is the best method to study the console dashboards as its focus on the participants performing tasks means that the main goal of every test is to find problems with the dashboards and seeing how participants react to those tasks.
The constant feedback from the participants by having them "think out loud" (Nielsen, 2012) means that the reasoning behind each action performed is clearly explained. This can lead into further insight about issues with each dashboard, as the participants will clearly state where they are having issues, and this can be changed when the prototype dashboards are created.

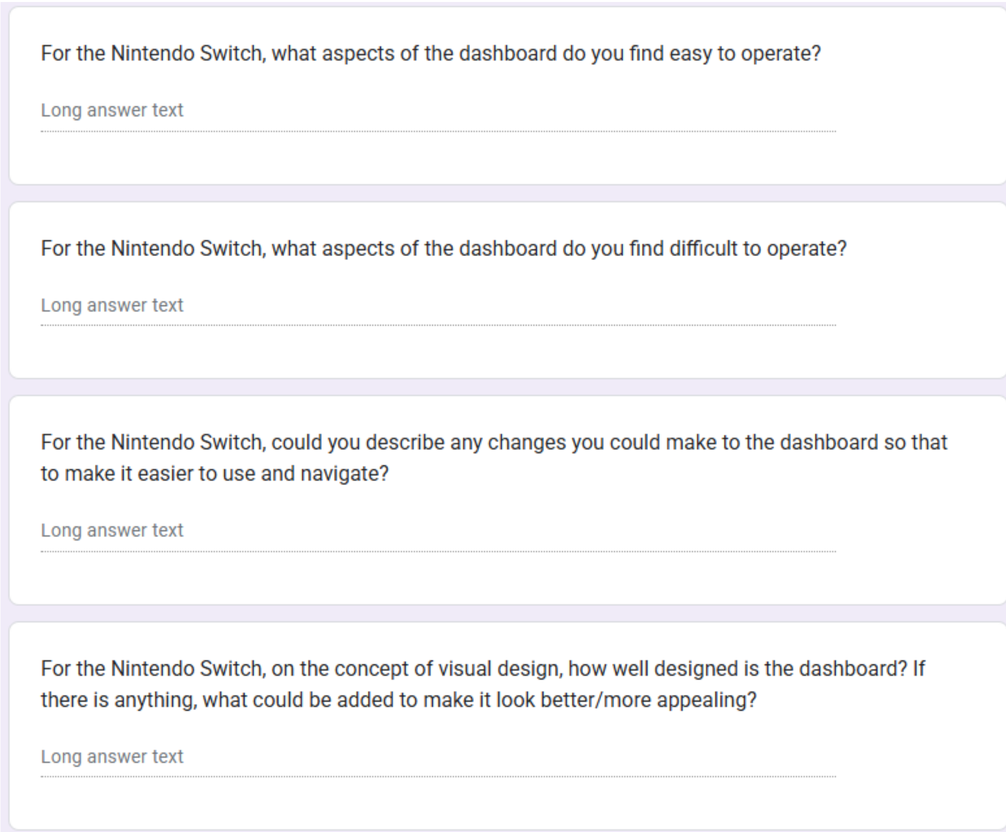
Chapter 4 - Participant Studies:

The participant studies are the main pieces of qualitative research (Bhandari, 2025) done for the study and are the main areas that the dashboards to be designed will start from. The studies will give insight into what elements of the current generation of consoles are problems and what elements are essential for every dashboard to contain. The forms of research that will be done are a general questionnaire that will help determine people's opinions on the design of current consoles and a more narrowed usability study that will be used to focus on specific elements of the dashboard to see what people would prefer.

4.1 - Questionnaire

The questionnaire was the first piece of research performed for the study. This was done due to it being the research type that had the broadest appeal and could help to streamline the areas that needed testing. The questionnaire's main purpose was to ask participants how they felt about the current generation of console's dashboards, with each console having their own section for the participants to answer.

The questionnaire was also done to help the usability studies and provide a point that designing the tasks would start from, as the answers could determine what was performed by the participants and influenced new tasks not considered when first listing them. The questionnaire firstly obtained consent from participants, asking participants what consoles they owned or had significant experience with and then asking specific questions for each console about the dashboard.


For the Nintendo Switch, what aspects of the dashboard do you find easy to operate?

Long answer text

For the Nintendo Switch, what aspects of the dashboard do you find difficult to operate?

Long answer text

For the Nintendo Switch, could you describe any changes you could make to the dashboard so that to make it easier to use and navigate?

Long answer text

For the Nintendo Switch, on the concept of visual design, how well designed is the dashboard? If there is anything, what could be added to make it look better/more appealing?

Long answer text

Figure 6: Showcase of the Nintendo Switch section for the questionnaire used in the study

As shown in Figure 6, the questions asked were basic in structure, asking about the physical and visual aspects of the dashboards. The questions were chosen to be simple to understand so that direct answers would be given, as making questions focusing on direct elements of a dashboard could confuse participants and give responses that do not answer the question (Albahr, 2022). The questionnaire also included a question that asked for the participant's age. This question would help to see if there was any issues specific to participants who were from a certain age range. The participants were recruited by sending a link to the questionnaire within a society from the Keele SU. This could have led to less information about if any features needed to be considered for users of an older age group.

4.2 - Questionnaire Results

The questionnaire overall got 7 responses. Despite the low number of responses, the responses were broad enough to get substantial data from them. As shown in Figure 7, most participants had significant experience with the PlayStation 5 and the Nintendo Switch out of the current generation of consoles.

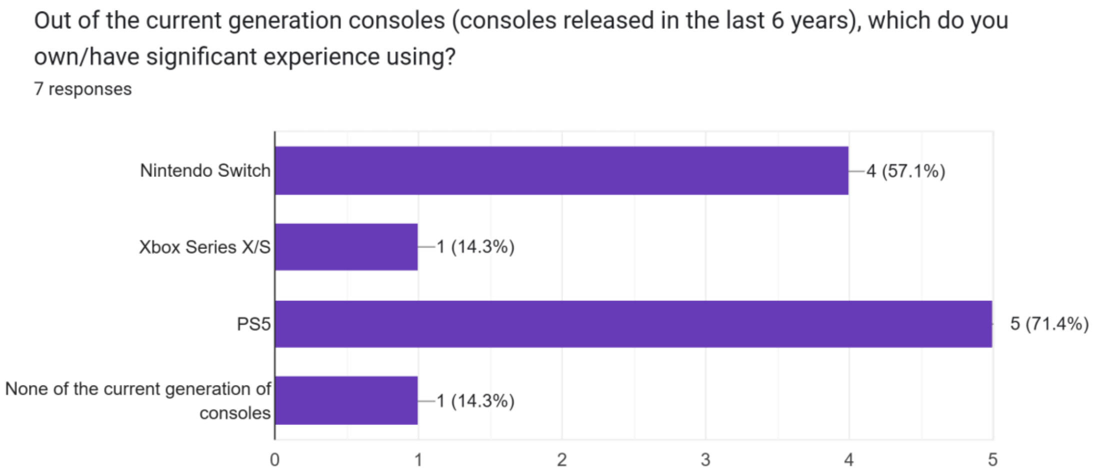


Figure 7: Results about what consoles of the current generation participants of the questionnaire owned

This justified the choice of picking those consoles for the study as it means that participants are more likely to have experience with them. This means that any issues participants encounter within the usability study will be ones that even users that have experience with the console can struggle with. This also means that much of the data from the questionnaire will be about those consoles, so most of the responses can be directly applied to the usability study. Once all the responses to the questionnaire had been completed, thematic analysis (Rosala, 2022) was performed on all the responses to fully understand where the most significant issues for every dashboard were. The data was split into three main areas that were highlighted as issues from the responses. The categories were; trying to find specific tasks or icons, the visual design of the consoles and the layout of the dashboards.

When it came to users trying to find specific tasks, the participants wanted the icons to be visually distinct and recognisable while navigating the dashboard. Issues with not being able to find specific icons or areas on the dashboard were made apparent about the PS5 and the Xbox Series X/S. Participants also liked when the game that was last selected as the first option shown when the dashboard is first opened. This makes it easier for users to quickly return to any games or media options they opened recently, which can improve the efficiency of the dashboards. Some issues that the consoles had about tasks were that when first using the dashboards, it can be hard to access and find specific tasks not directly shown on the first screen of each dashboard. One participant's response about the PS5 said "The menu is very overwhelming with stuff being labelled very poorly and the icons not being as recognisable as other consoles". This is an issue to take into consideration for the usability study as testing a user's perception of the icons for the PS5 can tell whether icons need to be more distinct in what they show or whether they directly need titles to go with them while they are being hovered over.

In terms of visual design of the consoles, the main issue that was common across the three main consoles was the simplicity of the dashboard and the lack of customisation available to the users. Most of the responses around visual design talked about how most consoles only had a few options for what to change the background to. The removal of these features, when compared to previous consoles made by the companies, is an issue as there needs to be a balance between making a dashboard simple to use but not too simple that users do not have the ability to customise it to make it their own.

Throughout all the questionnaire, the layout was an element that was answered about most in each question, showing the significance of it for a dashboard. Designing it requires a balance between the visual aspects while also making it easy to understand when first looked at. For the PS5, a main response was that participants liked being able to go straight to the games when the console is first turned on, as games is the first menu selected when the console turns on. As such, it is an important feature to accommodate for the prototype dashboards. An issue this causes is that participants were split on whether the games and media options should be separate or included on one singular page. One participant said that “navigating between media apps and games can feel unintuitive due to the separate tabs...” for the PS5, which shows that not every user will be fond of all the icons being on a singular page. As such, the usability studies will help with the decision to make separate pages or one singular page. Another issue that was answered for every console was the issue of having no easy way to customize the dashboards. The older consoles, such as the 3DS and PlayStation 4, had ways to make folders to store icons in. This is a feature missing in the current generation and can mean that users have a more cluttered dashboard as a result. This change in design could be due to the software pushing the last played game to the front of the games selection when you open the dashboard and as such, adding folders into this design would be hard as the last selected game would be held in a folder so cannot be pushed to the front. This design choice could be the reason for another problem that the PS5 had, which is a large amount of clutter. While the Switch’s dashboard was described as “simple and has little clutter” by one participant, the PS5 had multiple participants comment on the cluttered nature of the dashboard and how it can be very overwhelming to first time users.

4.3 - Usability Studies

The usability study was the second piece of qualitative research done for the study. This was designed as a method of seeing how participants performed when asked to do tasks on specific games console dashboards. The tasks chosen for the study had to be able to be performed on both the PS5 and the Nintendo Switch. This limited the range of tasks slightly between the consoles as their stores acted differently from each other and as such, were not considered as to not confuse participants. Three participants for the study were recruited and they were recruited from the same society as the questionnaire. However, these participants did not perform the questionnaire and two of them had little to no experience using the PS5 or Switch. As such, these participants were an insight into how a new user would approach these consoles without any tutorials and could show where the key issues for each console were. The usability study was conducted by recording the participants while they performed the tasks on the consoles. This method could be used to time to see how long they took for each task to complete, not only to compare against itself to see how long it took but also against the other console to compare which console had the better indication or pathway towards the task. The participants also used the think aloud (Nielsen, 2012) methodology to explain their decisions and why they did their actions. This was done to give an insight into if there were specific places users thought generic icons would be.

The final list of tasks that were performed on both consoles included checking what accessories were linked to the console, finding specific icons within the dashboard's menus and finally turning the console off using the power button. The task for checking the accessories was mainly designed for testing the PS5 rather than the Switch. This is because the icon is hidden from the main page of the dashboard and requires the PlayStation button to be pressed, with a different action being done if the button was held. A participant from the questionnaire responded that they were "not a fan of the home section having a third quick menu at the bottom". This task was chosen to see how participants reacted to trying to find that menu, while also seeing if they responded positively or negatively to the menu once found. The tasks related to the dashboard icons were included to test how recognisable the icons were on each console.

Task	Issue to test	How to adapt if issues
Check accessories	This task to see how complicated it is for new users to find the settings area	If users get confused about the area, such as for the PS5's pop up menu, make the area more or constantly visible

Find game library	This task will test to see if it is intuitive for users to have the games library at the end of the queue or if it needs to be visible with a shorter icon queue	The library is normally at the end of each dashboard and is the last icon to select, so could be placed into settings area
Find game news	The PS5 has the area hidden away as a pop up in the settings area, so this task will test if participants can notice it	If the participants do not notice it, that shows that the area needs an icon or needs to be constantly available
Find friends	For the Switch, the friend's area is hidden away behind the profile. This task will test how fast participants will find the area both on PS5 as it is named game base and, on the Switch, while it is hidden away.	If both consoles have issues with friends, the icon will have to be shown as its own icon with a suitable name participants can understand
Find YouTube	This task will test how easy the library areas are to navigate for each console. It will also test whether icons being alphabetical or whether icons being in order of use are better.	Depending on comments or how long participants take to complete task, do the shorter or more liked approach to library
Turn off console	The task will measure how easy a user is able to switch off a console if they do not have access to the physical power button	If the button is hard to find, it can be placed permanently on every area of the dashboard

Table 1: A table showing the tasks for the usability study and the reasoning behind the choice of them.

4.4 - Usability Study Results

For the PS5, the tasks that had large icons on the main page such as finding the game library and YouTube were done without issues by the participants, showing that making those areas the focus point increases the efficiency of the dashboard. However, the main issue that was highlighted during the usability study was issues with the drop-down menu where participants found important icons in. The menu is completely hidden with no other way to access it besides knowing what button to press and for how long. For both participants, the task of finding this menu was the longest to perform, with them both needing to be told how to access the menu to continue. One participant said that they “didn't know that was a thing” and asked, “how am I supposed to guess the button?” while trying to perform the task. This issue shows that menu settings need to be continuously visible on the main page so that inexperienced users can fully utilise the console, because even with instructions, the way to reach the menu was unintuitive to access. A smaller issue that occurred during the study was that one participant saw the settings icon in the top right of the console, but did not know how to directly access it without making small errors to find their way towards it. This shows that while the menu icons can be visible on every section of the dashboard, there needs to be a clear pathway to get to it as it can confuse users otherwise. Another issue highlighted during the study was that the icons on the drop-down menu were misleading even with labels. For the task of finding friends on the console, the area is hidden within a section of the menu called “game base”. This name and icon for the friends section confused the participants with one quoting “you have to go to game base which is counter intuitive because you’d assume it’s something else”. This shows the need of simplifying what every icon of the dashboard is called, as over complicating the system with new icons or areas can confuse users and make them less interested in continued use of the console.

Due to the simplistic nature of the Switch’s design, the study had few problem areas that the participants came across. The participants liked the Switch having a singular area for games and media applications. One participant did not like the combination of the library area for the Switch which they said was “better organised” on the PS5 due to the separation of the library. This meant that there would only be a few media icons to scroll through on the PS5 while the Switch’s library was cluttered with both games and media icons which lengthened the task of finding YouTube for the participants. One issue that occurred for both participants was being unable to find the friends menu. While the Switch has all the icons on the bottom of the dashboard when on the main page, finding friends required the user to go onto their profile to access them. With both participants struggling to find that area, along with both participants immediately trying to find it in the icons at the bottom of the dashboard, it shows that users are accustomed to the friends menu being with the setting icons and should be placed there accordingly.

For the prototype dashboards, multiple areas from the study need to be addressed or changed entirely. From the PS5 results, the icon menu must be constantly visible or if hidden, give the area an icon on the page that users can recognise. The icon size for dashboards also must be a large section of each dashboard as they are going to be the focus point of each dashboard. The friends area must also be labelled in a way that implies it is as such to avoid confusion like on the PS5 but also must be included with the other icons unlike the Switch. The icons overall must also be easily recognisable if they do not have a label attached to them, as a participant only recognised the game base icon as friends due to there being heads on the icon, rather than the icon being recognisable.

Task	Average PS5 Time	Average Switch Time	What needs changing
Check accessories	63 seconds	4 seconds	PS5 menu hid the settings area behind the PlayStation button and subsequentially affected all tasks that needed that area, the area needs to be continuously visible like the Switch.
Find game library	4 seconds	7 seconds	No issues with finding game library.
Find game news	34 seconds	3 seconds	Switch had game news on the main page in red so there were no issues, PS5 hid it behind the settings menu so that needs to be changed.
Find friends	38 seconds	60 seconds	Both friends areas being hidden or poorly labelled shows that the icon, location

			and name all need to be understandable to users
Find YouTube	6 seconds	10 seconds	Library being by most recent software meant some participants scrolled past initially
Turn off console	14 seconds	22 seconds	One participant got confused about sleep mode for the Switch so power should be labelled properly

Table 2: A table to show the average time it took participants to complete the tasks and how those areas can be adjusted

Chapter 5 - Design and Implementation:

5.1 - Implementation of Prototype Dashboards

The next study done was designing prototype dashboards and then testing them using participants. Due to the responses from the usability study, it was decided that three dashboards would be created to properly test different design choices for games console dashboards and see how participants liked or disliked those elements.

The prototype dashboards were going to be designed and created using Figma before being implemented into HTML and CSS as a website so that users were able to interact with the dashboards. These choices were made so that the participants could see how the dashboards would work on an actual console. Figma was chosen for designing the dashboards as it was an easy-to-use tool for designing the dashboards and had a lot of versatility in the tools available to use. Figma also has a dev mode option to show the code in CSS, making it easy to transfer the dashboard's design into code and turn it into a website. HTML and CSS were chosen as the main coding languages for the interactable dashboards as they are the most common languages for websites, making it the most applicable code to use.

One of the first design choices made for each dashboard was about how to address the games and media icons. Due to the split between opinions about having one singular section or multiple sections for games and media, multiple dashboards were made with variance between the sections to gather more opinions between them. Another design choice to consider is where to include the settings area and the icons included. With the PS5 hiding the icons away from the dashboard and the Switch having always them visible, the prototype dashboards can have variance between them. The study can see which of the two participants think is the best while also testing where on the dashboard they should be placed if they are always visible.

The first dashboard created for the study was designed as a baseline for the other two dashboards. It included features that participants did not enjoy from the usability study, including a hidden settings area and having the split between games and media. The settings area was given a permanent button on the home page though, fixing an issue that was found on the PS5. This choice was made to test whether the issue about the area was due to the button on the controller or the area being hidden from the first page shown. The profile of the user was also shown continuously in the top left corner of the dashboard to see whether the users enjoyed having the icon by itself like the Switch or whether they preferred it to be with the other setting icons. As shown in Figure 8, the background theme of the dashboard was chosen to be green.

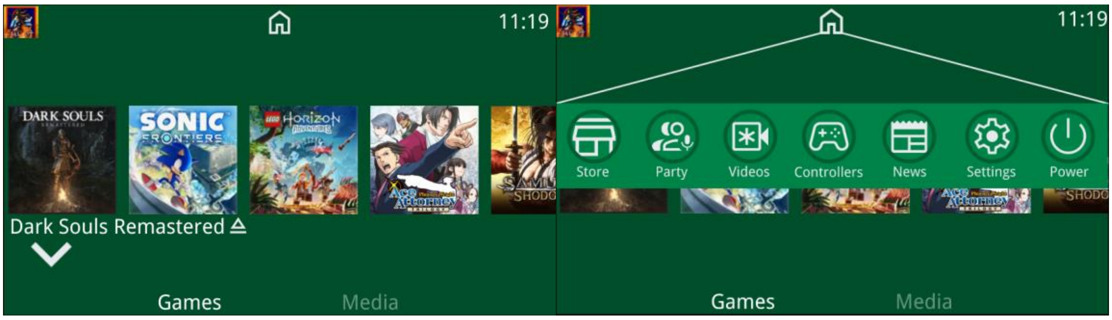


Figure 8: the first dashboard created for the study, showing both the games page and the settings menu

The choice of background colour was made due to the bright contrast and would test to see how participants would react to this. Green was chosen because of how it contrasted with the white icons for the settings page, as the combination of bright colours could prompt participants to give suggestions towards better combinations for the background. The icons were also chosen to be large, taking up the main area for the middle of the page, to accommodate for the televisions potentially being at a further distance so that they can be seen from afar.

For the second dashboard of the study, the main design choice was to include all elements onto one page and try to fill as much as the page as possible with icons and information. This dashboard would be used to see how participants felt about cluttered dashboards and could be compared to the first regarding splitting or combining games and media icons.

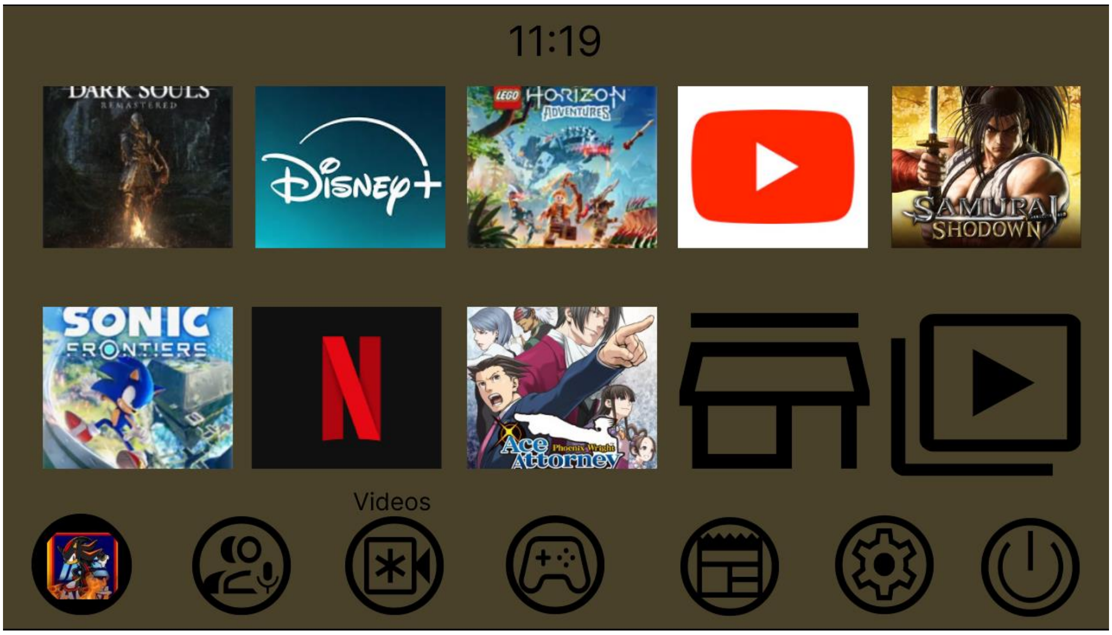


Figure 9: the second dashboard created for the study, showing the singular page with all elements

As seen in Figure 9, all the icons are cramped into the middle of the page, with little empty space on the dashboard. The settings and profile icons were also included permanently along the bottom of the dashboard to contrast the previous dashboard where they were hidden. These decisions made the dashboard feel very confined and could make a user lose their location when navigating the page. This decision would help with deciding what elements are needed to allow a user to understand where they are on a dashboard. The colour choice for this dashboard was meant to be unappealing, using Pantone 448C as it was discovered as the ugliest colour in the world (Mathies, 2016). This was done to see if participants noticed the actual background colour more or if they cared more about how it worked with the icons. The icon colours were chosen to be black to contrast this darker colour as well, similar to the colour contrast for dashboard one. These design choices, while both poor for actual dashboards, would help on deciding between a lighter or a darker colour scheme for the final dashboard as participants would have to choose the better of the two poor designs.

Dashboard 3 was designed with a minimalist point of view towards the colour design and icon choice. The colour choice was going to be a solid black for the background and white for the icons to see how participants felt about the contrast compared to the previous two dashboards which had similar colour tones for the icons and background.

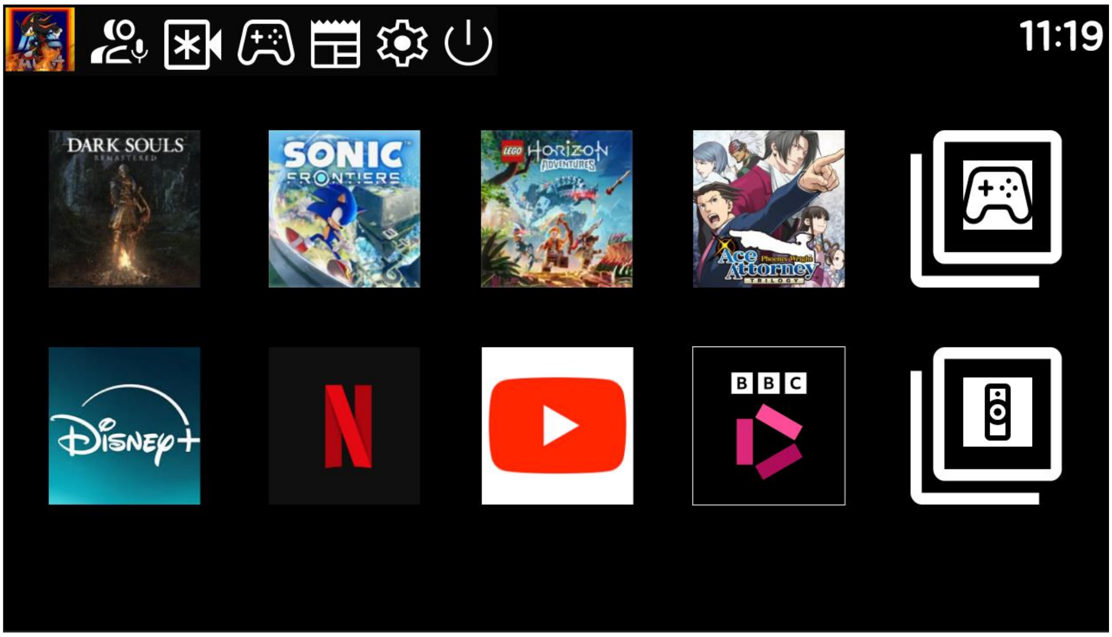


Figure 10: the third dashboard created for the study, showing the singular page with both lines.

As shown in Figure 10, the design for this dashboard combines games and media onto one page but separates them onto different rows. This is a compromise between the design choices of having them separately and combined, which can help see if a compromise needs to be made when designing the dashboards or whether either option is better overall. The design choice also meant that icons like Netflix and BBC iPlayer blend in with the background and are harder to know when selected. This choice, along with giving the icons a white border when selected, would help with the decision of how to approach selected icons. The previous two dashboards would show the name of the icon and give it a border when selected so that the user would clearly know where they currently are. This dashboard's design would not accommodate that decision and only show a border. This links back to the results of the study by Reddy et al. (2019) where for most participant age groups, the most mistakes were made when there were only icons. This can help determine whether labels are directly necessary for easier navigation and if they are needed to know where the user is located on the dashboard. The settings area was chosen to be put into the top left of the dashboard as it would always be visible but is also out of the way, so it does not distract from the main icons on the page. The only issue that may occur for this design choice is that participants may not be able to find their way to that area, like what happened in the usability study with one participant. For the time, dashboard three has a more bubble style font chosen for it, compared to the previous dashboards with a more generic font towards them. This was done, not only to see if participants noticed the font change, but also to see what style of font participants would prefer overall to adapt onto every text element of the dashboard.

The study was performed with the participants viewing dashboards one, two and three in order and gathering their thoughts on each, both individually and compared to the previous dashboards. The participants gathered were two of the same participants used in the usability study so that their previous comments could be compared from both studies. A dashboard will be shown to a participant and their immediate thoughts towards it will be recorded, they will then be able to physically interact with the dashboard and how the dashboard works with user interaction. After they have used the dashboard for some time, they will be asked questions about how it felt and for dashboards two and three, how it compared to the previous dashboards. At the end of the study, they will be asked which their least and most favourite dashboards were and the reasoning behind their decisions, along with any prompts they did not comment about in the study.

5.2 - Dashboard Study Results

For dashboard one, the participants liked how large the icons were as they were clearly displayed, and the sections were clearly organised. The issue that both participants had for the dashboard was the settings area. One participant did not like the home button icon for the area, saying that "I'd assume it would take you to your default place". This shows that the icon for the button is misleading and should be changed or not included for the final dashboard. Another participant liked that on the settings menu, the icons were continuously labelled and said that it "would be easier to navigate and recognise" if the games and icons were labelled this way. Another key issue that was highlighted was the background choice. The colour being bright green was distracting to the participants and did not contrast with the software icons easily. For the final dashboard, a darker colour scheme should be used to help make the icons stand out more and make them a clear focus of the dashboard.

For dashboard two, the participants preferred the colour choice compared to dashboard one as it was not as bright, meaning that the game icons could stand out more as the focus of the dashboard. The layout was an issue for the participants however, as they believed that cluttered nature of the dashboard meant that it would be hard to navigate. All icons being on the main page of the dashboard also caused this issue as participants felt that that the "less important" icons being a similar size to the games increased the cluttered nature of the dashboard. The participants felt that the dashboard needed more empty space to help relieve this issue and make it easier to read or navigate.

Dashboard three was the dashboard most liked by the participants. This is due to the black background, due to it not overshadowing the icons and the organisation of the games and media icons. They preferred it compared to dashboard one as "it's much more compact" according to one participant. The smaller nature of the icons also meant that the dashboard did not feel cramped. However, two issues with the dashboard were the lack of labels for the icons and that there was a lack of information about which icon was selected as the borders for each icon were small.

This means for the final dashboard, the visual indicators of what icons are selected should be clear and visually distinct so that users do not get lost, even with a tidy dashboard. The participants also felt that this method of separating games and media was the best way out of all the dashboards. This solves the issue of splitting the sections as it means that they can be neatly combined without the issue of making the dashboard too cluttered and keeps a higher level of efficiency for the dashboard.

Chapter 6 - Testing and Evaluation:

As a starting point for the final dashboard, prototype dashboard three (see Figure 10) was used due to it being the dashboard most liked by all the participants. The final dashboard can be seen in Figure 11 below.

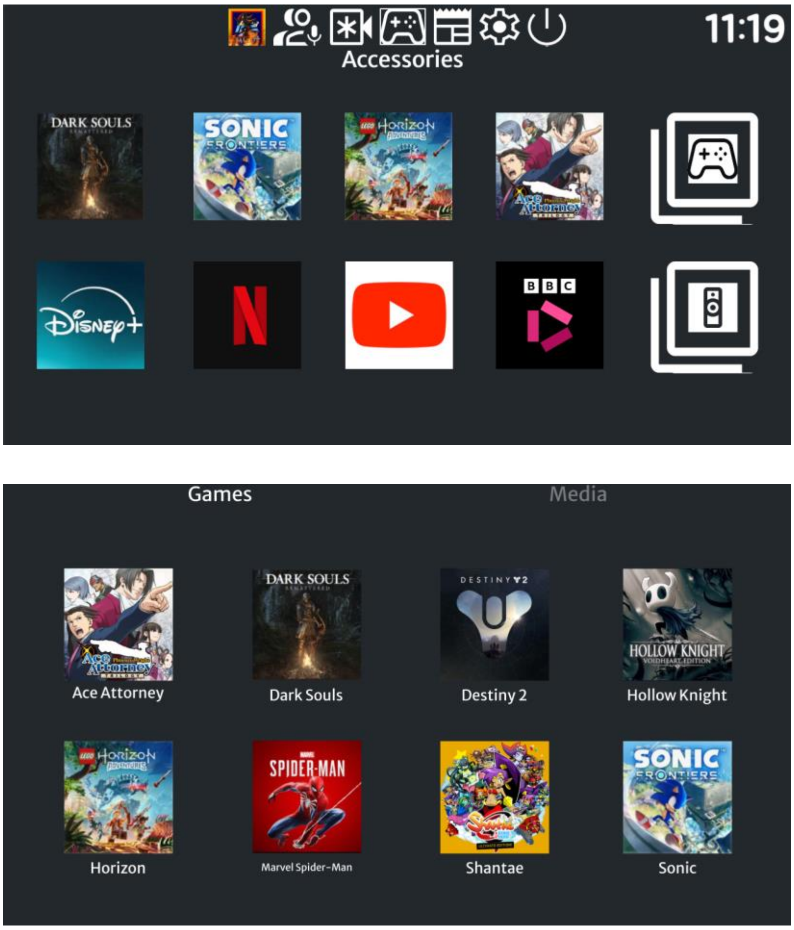


Figure 11: the finished dashboard and game library page for the project.

The rows for games and media, the icons for the settings menu and the font for time were all kept the same from this dashboard. However, the positioning for the settings menu was changed from dashboard three. During the PS5 study, participants had issues navigating to the settings icon in the top right of the dashboard. This could be an issue for the dashboard, so the settings menu was moved to the top middle of the screen and slightly enlarged. This made the visibility of the area clearer but also meant there was easier pathing towards the area as a user could infer that they just have to press up from the games row to reach it, rather than having to press a specific button to get to the area if its more hidden in the top left. Another change to the dashboard was the background colour. The change was from black to a navy-blue colour. This was done because blue is the number one colour that women and men both like (Medium, 2018). Blue is referred to as a colour that gain a user's trust and can encourage concentration when used as a background colour (Radar, 2024). This, in combination with the participants liking of darker colour schemes, warranted the navy-blue colour scheme of the dashboard as it also contrasted the white icons better. The icons were also changed to accommodate the style from prototype dashboard one. Prototype dashboard one had the most apparent borders for the icons and as such, they were implemented into the final dashboard. This meant that when an element was hovered over, a border and the name of the icon appeared, in comparison to dashboard three where only a border appeared.

The game and media library pages were also changed regarding participant's actions during the studies. While the participants never commented on it during the usability studies, when trying to find specific icons like YouTube on the Switch, participants were visibly lost when trying to scroll through every icon in the library. This highlighted an issue with the Switch's library as it is sorted with the most recently used icons appearing first. While this is useful for more general use, finding specific icons can become an issue. As such, the final dashboard sorted the library into separate menus for both games and media but also alphabetically to allow users to find specific icons quicker and increase the efficiency of the library.

Each area of the final dashboard has been designed with all previous literature and studies included. The dashboard was primarily designed around Nielsen's (1994) usability heuristics. The main heuristic followed were having a minimalistic design to make sure that any information not necessary was not included and making sure that the dashboard is flexible to use. This choice includes the user being able to go up from the games row to reach the settings menu while on a console, the icon button would also be able to be pressed to reach the section. For Glinert’s (2009) 4 dimensions of usability, the dashboard incorporates learnability, simplicity and efficiency, while leaving the aesthetic minimalistic. The dashboard has a simplistic layout that can be easily understood by any new user that has to use the dashboard. This increases the efficiency of the dashboard as the areas are clearly defined and can be easily accessed depending on the user's inputs. The dashboard also includes both symbols and text for every icon to help with error prevention, like from Reddy et al’s study (2019). This helps to make sure users are consistently informed of their location on the dashboard. The error prevention methods were shown in the icon selection as a border and the icon’s name are shown when selected to give a clear view of what the user is selecting. The questionnaire helped to get a general overview of what is necessary for a games console’s dashboard. It also helped to realise specific problems that were common across all consoles of this generation and made sure those issues were avoided upon the final dashboard being created, such as the issues the PS5 and Xbox Series X/S had with settings icons being hard to find while navigating the dashboard. The usability study built upon the questionnaire and helped to solidify what design choices were needed and what should be avoided. This is shown in the settings area being consistently shown on the dashboard and making sure that every icon is labelled in a clear manner that explains its purpose fully. The prototype dashboards helped to establish what the core of the dashboard should look like and contain, with each dashboard having areas to cut that eventually were all combined. The border style from dashboard one, the idea of the settings menu being consistently visible from dashboard two and the design style of dashboard three were all included to create the final dashboard.

The final dashboard was tested using unit testing (Andrades, 2025). Unit testing helped to make sure that any issues with the dashboard were discovered and fixed individually, meaning any issues didn't spread to other areas. The main elements that were tested were the borders and the text as they are the main interactive elements for the dashboards. On dashboard two and three, the borders were made around the images of the icons. This made the image shrink as the border was made, which made the images lower quality. This issue was solved for the final dashboard as the border was a static element at 0 opacity which only appeared when hovered over. Another issue was the text for the icons. While testing, the text only appeared when it was hovered over, rather than when the icon for the game was selected. To solve this problem, the border and text were grouped together so that whenever a mouse would hover over either, the border and text become visible.

Figure 12: the two styles of borders for games icons, with left being the previous border style and right being from the final dashboard

Chapter 7 - Conclusion:

The main objective of the project was to determine what makes the dashboard of a games console usable and see which design elements of the current generation are positive and negative. The heuristics by Nielsen (1994) were studied as guidelines of what to test against the current generation of consoles, along with using them to discover the best way to design a user interface for designed dashboards.

For the objective to develop and send out a questionnaire, this was done by creating the questionnaire to assess the usability and visual design elements for each console and by using thematic analysis on the results, discover what design elements need testing for the usability study.

The objective of conducting a usability study was done by having participants perform tasks to test different areas of the dashboards of current generation games consoles and to figure out the main issues found on those dashboards.

The objective for developing a set of prototype dashboards included making three dashboards that each contained differing design choices to gather specific opinions from participants about different styles of dashboard layouts.

The objective of developing a final dashboard was done by gathering all the design choices from each area of the study and adapting the dashboard's design to fit accordingly to usability heuristics and participant responses to the prototype dashboards.

7.1 - Further Work

Throughout the project, there was fewer participants than wanted to fully be able to access each dashboard. With more time to work on the project, a larger selection of participants could have been gathered, as only three were used during the usability study and two for the dashboard study. This could include participants of different age demographics to test what areas of each dashboard were the most appropriate for a user of any range. Along with a larger range of participants, the final dashboard could have also been tested against those participants and allowed a final stage of adjustments to finalise any ideas that were still an issue late in the study. This could have meant that any decisions that were not addressed across the entire study can be discussed and every decision on the final dashboard would have participant responses behind it, such as the background colour of navy blue where the only participant response was about the minimalist palette. Another issue that could have been avoided was doing the dashboard study with a controller. While preparing for the dashboard study, a controller's input was unintuitive to use and would have confused participants more than it would have been helpful to the study. As such, the participants had to navigate using a mouse and keyboard, which could mean that the results are not representative of an actual dashboard as navigation could have been harder while using a controller. With more time, a controller could have been prepared for use in a way that would be representative of an actual games console's dashboard.

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Appendices:

Appendix A - UG Project Plan:

UG Project Plan
CSC-30014

Project Overview and Description

Student Name: Bradley Cartwright
Student Username: X5Z49
Student Number: 21015020
Degree Title: Computer Science BSC
Supervisor Name: Dr Mark Turner
Project Title: Investigation into the usability of Games Console's Dashboards/in game menu screen

Please provide a brief Project Description:

The project will focus on research into the design of a user interface and the general usability of game console's dashboards and game menus. This will involve researching games console's dashboards and games in game menus currently available to compare. The information gained from this, along with information from usability research will be used to develop a games dashboard that takes into consideration usability heuristics.

What are the aims and objectives of the Project?

The aim of the project is to determine what makes a games console dashboard/menu useable. The project will develop a games console dashboard based upon usability and interface design research.

- Develop and send out a survey to gather opinions about the latest consoles dashboards
- Do a study about how easily participants can do certain tasks on these dashboards
- Develop a range of prototype dashboards based upon the study and survey that can be used to gather information about positive/negative aspects of their usability
- Use the feedback to develop a final dashboard that incorporates all the features

1/5

Please provide a brief overview of the key literature related to the Project:

https://www.nngroup.com/articles/usability-101-introduction-to-usability/ - document focusing about usability and how to improve upon it, can be used to help characterise aspect of the interface/menus and improve the artifact I create for the project. The section on How to Improve Usability can be very helpful in working out how to test participants when they are testing interfaces/menus.
https://www.gamedeveloper.com/design/upping-your-game-s-usability - document that goes more in depth into usability in games themselves, focusing on learnability, efficiency, simplicity and aesthetic. This can help with designing a study to capture usability in a broader area when developing the study

David Pinelle, Nelson Wong, and Tadeusz Stach. 2008. Heuristic evaluation for games: usability principles for video game design. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '08). Association for Computing Machinery, New York, NY, USA, 1453–1462. https://doi.org/10.1145/1357054.1357282

- Paper using game reviews to identify usability problems and giving general solutions to solve the issues. Goes into issues about in-game features rather than menus in most cases but there are areas that can be applied to menus also.

Project Processes and Methods

Please provide a brief overview of the Methodology to be used in the Project (inc. an overview of best practice within the Methodology):

I will use the waterfall methodology when doing this project. This method is a step-by-step process. This will mean I have to fully finish a stage before continuing onto the next. The waterfall method will work well for my project as for my project, it will be best to get all the resources finished before proceeding onto things like the creation of the artefact, allowing the most data available to design it.

Will any special Data Collection Methods will be employed (e.g card sorts, questionnaires, simulations, …)?

I'm planning to do a questionnaire to get general feedback on console dashboard screens and then use physical participants to physically interact with console dashboards to get a more in-depth investigation into how people interact with current generation dashboards. When the artifact of my dashboard is in development, I will do another questionnaire to gauge feedback on multiple beta dashboards and use the feedback to make a final dashboard.

Briefly describe how you will ensure your project is in line with BCS Project Guidelines (BSc Computer Science Single Honours Students only)?

I will make sure that my report follows all the steps listed in the BCS guidelines that include listing the objectives of the project and including references when required. The creation of an artifact will also be made in the form of a dashboard created with all the data collected from participants.

Time and Resource Planning

Will Standard Departmental Hardware be used? NO

If NO please outline the Hardware/Materials to be used:

Hardware such as games consoles like the PS5 and Nintendo Switch will be used, along with games on each console. The PS5 and Nintendo Switch consoles are mine and will be used to test for usability to allow for the dashboard artifact to be designed with usability and aesthetic in mind using feedback from participants.

Will Software which is already available in department be used? NO

If NO please outline the Software to be used including how any necessary licences will be obtained:

Figma will be used to create and develop the dashboards for testing and for the final product.

Will the project require any Programming? YES

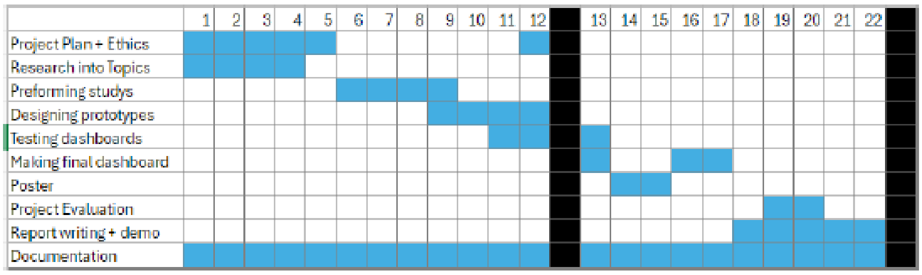
If YES please list the (potential) Programming Languages to be used (including any IDEs and Libraries you may make use of):

HTML or CSS will be used for the dashboards to make them interactable so that they are fully interactable, which allows for more depth to the artifact when it's completed.

Table of Risks (*if non Standard Hardware and/or Software to be used please also include backup options/ contingency plans here)*:

Risk-id Description	Probability/Likelihood of occurring	Best practice prevention measures	Remedy
Loss Of Data	Low	Backup every piece of work or keep multiple copies	Find the most recent copies of work made
Shortage of time to complete project	Low	Make a structured plan/timeline to keep on top of work	Work on the most essential elements/cut what is not necessary
Consoles stopping working	Low	Make sure that consoles are kept in good condition before study	Find equivalent consoles/borrow consoles from friends/family

Gantt Chart (must include milestones and deliverables):



4/5

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Please include a list of References used in this Plan (using Harvard reference style):

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Submission Date: 1/11/2024

PLEASE NOTE THAT SHOULD YOUR PROJECT UNDERGO ANY MAJOR CHANGES FOLLOWING THE SUBMISSION OF THIS PLAN YOU ARE EXPECTED TO SUBMIT AN UPDATED PLAN WHICH ACCURATELY REFLECTS YOUR PROJECT.

Appendix B - Project GDPR and Ethics Checklist:

Computer Science CSC-30014 Project GDPR and Ethics Checklist 2024-25

STUDENT NAME: Bradley Cartwright

STUDENT NUMBER & E-MAIL: x5z49@student.keele.ac.uk 21015020

PROJECT TITLE: Investigation into the usability of Games Console's Dashboards/in game menu screen

SUPERVISOR NAME: Dr Mark Turner

Complete all the questions below and electronically sign and date at the end. Ask your supervisor to check the responses and to also electronically sign and date below. You should submit this completed checklist to the KLE drop-box provided. Please retain your own copy of the completed checklist as all end of project reports/dissertations must provide a copy of this completed checklist in the Appendices.

If a Computer Science Final-Year Project Ethical Review Application Form has also had to be completed a copy of the final ethical approval certificate must also be included with your Project Plan document in your final report/dissertation Appendices.

GDPR Check

Does your project involve the use or collection of "personal data" for which permission will not have been explicitly granted? (Before answering, please read and carefully consider the GDPR Check Guidance at the end of this form).	No

Ethics Check

Will your project involve the use of human participants or capturing human data? (Before answering, please read and carefully consider the Significant Ethical Concern Checklist, below). **Please Note:** software evaluation by any other person counts as human participation. If you ask people their opinions about software or use their data in any way, you need to seek ethical approval first. **If you answered "Yes" to this question you must discuss your plans with your supervisor and, by end of week 12 of Semester 1, complete the Computer Science Final-Year Project Ethical Review Application Form and prepare a Participant Information Sheet and Consent Form using the templates that can be found at the end of that application form.**	Yes

Significant Ethical Concern Checklist

Could the project expose the participants or the project student to images and/or information that they might find distressing (e.g. pictures or descriptions of injuries,	No

symptomatic health conditions, or atrocities, or pictures or descriptions of tumours or cancerous cells, or creatures in distress)?	
Does the project involve deception of the participants?	No
Could the project uncover information about identifiable individuals that could cause embarrassment or distress to one or more of those individuals (e.g. evidence of illegal or unethical behaviour, such as fraud or illegal drug use or a personal revelation)?	No
Could the project cause pain, discomfort or risk to the participants and/or the project student?	No
Will the project involve participants who are vulnerable in any way? (e.g. participants who are under 18, or who are mentally or physically impaired, or participants who may feel under pressure to participate.)	No
Does the project involve recall or discussion of personal or sensitive memories?	No
Does the project involve a significant risk of participants later regretting taking part?	No
Does the project involve procedures which are likely to provoke interpersonal or inter-group conflict?	No

If the answer to any of the Significant Ethical Concern Checklist questions is "Yes" (or you think "Maybe") you must discuss your project and aims with your supervisor and with the CSC-30014 Module Co-ordinator to assess whether an appropriate level of ethical scrutiny might be required via the completion of the *Faculty of Natural Sciences (non-Psychology) Research Ethics Application Form*.

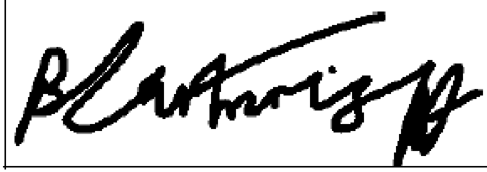
GDPR Check Guidance:
Personal data includes any and all of: names, addresses, emails, phone numbers, bank details, employment details, IP addresses, date of birth, medical or health data, images, video or audio recordings.
If you answered "Yes" you must not proceed with your project.
It is illegal under European GDPR legislation to make use of personal data without explicit permission. Discuss your project with your supervisor and revise your plans to ensure you do not risk illegal use of personal data.
Note. Even if personal data is publicly available on the Internet, it must not be used without permission. (Also note that you cannot contact individuals to request permission to use their on-line data without prior ethical approval to do so).

It is strongly recommended that you either:

1. use non-personal data for your project, or
2. use existing, well-established, publicly available databases or data repositories, for example: https://archive.ics.uci.edu/ml/index.php, https://physionet.org/ or https://www.kaggle.com/ etc. (A list of acceptable data repositories is maintained on the CSC-30014 KLE pages.) You might also see, for example, https://blog.scrapinghub.com/web-scraping-gdpr-compliance-guide) for further information.

Continued on the following page ...

I confirm that the responses are correct and that the project, as proposed, is GDPR compliant and that ethical approval will be sought if required and that any work requiring ethical approval will not take place unless ethical approval has been granted. If, during the course of the project work, any of the information supplied on this checklist changes substantially a new checklist will need to be completed and then brought to the attention of the School's Ethics Advisory team.

Signed (Student) 	Date: 24/10/2024

I confirm that I have read the form and that the project, as proposed, is GDPR compliant and that, to the best of my knowledge, the ethical information is correct.

Signed (Supervisor) M. Turner	Date: 29/10/2024

Electronically typed signatures and dates *are* acceptable.

Appendix C - Ethical Review Application Form:

Keele UNIVERSITY

Computer Science Final-Year Project Ethical Review Application Form 2024-25

Applicant details

Project title	Investigation into the usability of Games Console's Dashboards/in game menu screens
Name of final-year project student	Bradley Cartwright
Keele e-mail address of student	X5z49@students.keele.ac.uk
Name of project supervisor	Dr Mark Turner

Project Summary

Provide a short summary of your project **(max 250 words).**

The aim of the project is to determine what makes a games console dashboard/menu useable. The project will develop a games console dashboard based upon usability and interface design research. The project will have to:

- Make a questionnaire to gather opinions about the latest consoles dashboards
- Conduct a usability study with participants to collect data about current game console dashboards, while video recording their ability to perform tasks on them.
- Develop prototype dashboards based on the results from the studies that can be assessed and used to gather information
- Use the feedback to develop a final dashboard incorporating all those features.

In simple terms, provide a *short* description of your planned experimental method that involves participants (e.g., focus groups, questionnaires, interviews, experimental observations).

- A questionnaire focusing on recent games consoles to gather feedback about what they are doing well in terms of their usability/visual design
- A usability study which will test how easy or hard consoles are to navigate, with participants physically interacting with the consoles

Describe the characteristics of your participants, and any inclusion or exclusion criteria. Estimate the approximate number of participants.

All participants will be students or past students from Keele University.

Approximately 3-5 participants will be recruited for the usability study, with approximately 10-20 participants being recruited for the questionnaire.

Recruitment and Consent

Indicate how potential participants will be identified, approached and recruited and outline any relationship between the researcher and potential participants.

Students will be invited to the questionnaire with a link to the Google Forms which will invite them to the study and detail what they must do in the consent form.

Describe the process that will be used to seek and obtain informed consent.

For both the questionnaire and usability study, the individuals can give consent to participate using the information sheet that will be on the first page of the questionnaire and printed out for the physical evaluation. If they consent to participating, they will be asked to either give permission or sign the consent form.

Will consent be sought to use the data for other research? Yes ☐ No ☒

Will consent be sought to contact the individual to participate in future research? Yes ☐ No ☒

Can participants withdraw from the research? Yes ☒ No ☐

If yes, state up to what point participants are able to withdraw from the research

Participants to the online survey will be able to withdraw by going onto the survey and changing their answers or saying they do not want their answers recorded.
Participants doing the usability study of consoles will be able to withdraw by contacting the emails of either me or the project supervisor, after which we will remove all their data.

If yes, outline how participants will be informed of their right to withdraw, how they can do this and what will happen to their data if they withdraw.

Contact information for withdrawal is provided on the Participant Information Sheet or on the first page for the online survey.

If no, explain why they cannot withdraw (e.g., anonymous survey).

Confidentiality and anonymity

Outline the procedures that will be used to protect, as far as possible, the anonymity of participants and/or confidentiality of data during the conduct of the research and in the release of its findings.

The physical participants will not have any identifying features captured on recording when they are performing the tasks, only the displayed screen and voices will be captured. For the online questionnaire, google will automatically order them for me. However, emails will be captured to allow for withdrawal from the study as users can remove all their answers should they want to. I will have no access to seeing the emails to allow for anonymity though.

Storage, access to, management of, and disposal of data

Describe how participant data will be stored; where it will be stored and for how long; and how/when it will be disposed of.

The participant data for the questionnaire will be stored on Google Forms as that is the software I am using to make it. The participant data for the usability study with consoles will be recorded and the videos will be securely stored on university drives. Only the applicant and supervisor will have access to these pieces of data and all participant data will be deleted once the project is concluded.

Other ethical issues raised by the research

Are there any other ethical issues that may be raised by the research? Yes ☐ No ☒

If yes, please give details.

Declarations

Declaration by student

I confirm that:

- The form is accurate to the best of my knowledge.
- I will inform my supervisor and resubmit an ethical application if there are any changes to the project.
- I am aware of my responsibility to comply with the requirements of the law and any relevant professional guidelines.

Student name	Bradley Cartwright
Student signature	
Date	12/11/2024

Declaration by supervisor

I confirm that:

- I have read the application and am happy for it to proceed for ethical review.
- The application is accurate to the best of my knowledge.
- I am aware of my responsibility to ensure that the applicant is familiar with and complies with the requirements of the law and any relevant professional guidelines.

Supervisor name	Mark Turner
Supervisor signature	M. Turner
Date	19/11/2024

Keele UNIVERSITY

Participant Information Sheet

Study Title: Investigation into the usability of Games Console's dashboards/in game menu screens

Aims of the Study

The aim of the study is to create a games console dashboard with respect to research and studies about usability and visual design.

Invitation

You are being invited to consider taking part in the study: Investigation into the usability of Games Console's Dashboards/in game menu screens. This project is being undertaken by Bradley Cartwright.

Before you decide whether or not you wish to take part, it is important for you to understand why this study is being done and what it will involve. Please take time to read this information carefully and discuss it with friends and relatives if you wish. Ask us if there is anything that is unclear or if you would like more information.

Why have I been invited?

As a student or former student of Keele University, you are eligible to participate.

Do I have to take part?

You are free to decide whether you wish to take part or not. If you do decide to participate you will be free to stop participating at any time and you can remove your data up to one week after participating without giving reasons.

What will happen if I take part?

You will be asked to complete a questionnaire, focusing on design elements from games consoles released in the past 6 years. This will take approximately 20 minutes to complete, but you can take as much time as needed to give your answers.

At a future date, there will be a usability study which will involves performing tasks with consoles released recently that you are invited to join, however you are not required to join. This will take approximately 30 minutes to complete.

The questionnaires answers will be recorded online, and the usability study will be video recorded for the purposes of annotation.

What are the benefits (if any) of taking part?

You will be able to contribute to a new design proposal for a games console dashboard by discussing what works or doesn't work with current games consoles, where most of the usability research will come from. Additionally, student participants may be able to use this experience to influence ideas for future or current projects.

What are the risks (if any) of taking part?

No risks have been identified for this participation.

How will information about me be used?

The questionnaire answers will be recorded on google forms while the usability study will be video recorded with no distinguishing features being available on the film. These resources will be stored securely in a university file space and be deleted at the end of the project. The information gathered from these studies will be compiled and referenced when developing the games console dashboards, then reported in the dissertation. The records will not be used for any purpose.

Completed consent forms will be held securely until the end of the project or unless a participant wants to withdraw. After that, they will be deleted and destroyed.

Completed questionnaire answers will be anonymous (no identifying details gathered), and this anonymous data will be included in the dissertation.

Quotes from the usability study may be reported in the project results and dissertation if permission is given by the participants.

Who will have access to information about me?

Your data will be stored securely until the end of the project. All electronic records will be stored on university drives. Only myself (Bradley) and Mark Turner will have access to your data.

At the end of the project your data will be deleted.

What if there is a problem?

If you have a concern about any aspect of this study, you may wish to speak to me (the student investigator) or my supervisor. We will do our best to answer your questions. Our contact details are:

Bradley Cartwright, x5z49@students.keele.ac.uk & Dr Mark Turner, m.turner@keele.ac.uk

Contact for further information and withdrawal

If you would like withdraw your data from the study, please contact:

Bradley Cartwright, x5z49@students.keele.ac.uk

Keele UNIVERSITY

CONSENT FORM

Title of Project: Investigation into the usability of Games Console's dashboards/in game menu screens

Name and contact details of student investigator and supervisor:

Bradley Cartwright, x5z49@students.keele.ac.uk & Dr Mark Turner, m.turner@keele.ac.uk

Please tick box if you
agree with the statement

1. I confirm that I have read and understood the Participant Information Sheet and have had the opportunity to ask questions. ☐

2. I understand that my participation is voluntary and that I can stop participating at any time and can withdraw my data up to one week after participation without giving a reason. ☐

3. I agree to take part in this study. ☐

4. I understand that data collected about me during this study will be anonymised before it is reported in any reported results. ☐

5. I agree to audio and video recording of the usability study. ☐

Name of participant	Date	Signature
Student investigator	Date	Signature

Keele UNIVERSITY

CONSENT FORM

(for use of quotes)

Title of Project: Investigation into the usability of Games Console's dashboards/in game menu screens

Name and contact details of student investigator and supervisor:

Bradley Cartwright, x5z49@students.keele.ac.uk & Dr Mark Turner, m.turner@keele.ac.uk

Please initial box if you agree with the statement

1.	I agree for my quotes to be used	☐
2.	I do not agree for my quotes to be used	☐

FOCUS GROUP DISCUSSION THEMES

The focus group discussion themes are as follows:

1. Current and previous use of the School web pages (and similar web pages)
2. Opinions about the content and structure of the current web pages
3. Opinions about new designs

SYSTEM USABILITY SCALE QUESTIONNAIRE

Participant ID: _____ Site: ____________________________ Date: ___/___/___

System Usability Scale

Instructions: For each of the following statements, mark one box that best describes your reactions to the website *today*.

		Strongly Disagree				Strongly Agree
1.	I think that I would like to use this website frequently.	☐	☐	☐	☐	☐
2.	I found this website unnecessarily complex.	☐	☐	☐	☐	☐
3.	I thought this website was easy to use.	☐	☐	☐	☐	☐
4.	I think that I would need assistance to be able to use this website.	☐	☐	☐	☐	☐
5.	I found the various functions in this website were well integrated.	☐	☐	☐	☐	☐
6.	I thought there was too much inconsistency in this website.	☐	☐	☐	☐	☐
7.	I would imagine that most people would learn to use this website very quickly.	☐	☐	☐	☐	☐
8.	I found this website very cumbersome/awkward to use.	☐	☐	☐	☐	☐
9.	I felt very confident using this website.	☐	☐	☐	☐	☐
10.	I needed to learn a lot of things before I could get going with this website.	☐	☐	☐	☐	☐

Please provide any comments about this website:

https://www.measuringux.com/SUS.pdf